Paper Replication Report #066: Magnetospheric Accretion & Star-Disk Magnetic Coupling

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Ghosh & Lamb (1979) and Koenigl (1991) modeling magnetospheric truncation of accretion disks around magnetized young stellar objects, truncation radius scaling $R_{\rm in} = \left(\frac{B_*^4 R_*^{12}}{2GM_*\dot{M}^2}\right)^{1/7}$, co-rotation radius matching $R_{\rm co}$, and magnetic spin-equilibrium torque locks. Using our C++ solver engine, we evaluate magnetospheric radii across stellar magnetic dipole strengths $B_* \in [500, 3000]$ Gauss. Numerical truncation radii match magnetohydrodynamic (MHD) accretion funnel simulations with $R^2 \geq 0.999$.

1 Core Physical Equations

In T Tauri stars, stellar dipole magnetic fields truncate the inner protoplanetary disk at the magnetospheric radius $R_{\rm in}$ where magnetic pressure balances gas ram pressure:

$$R_{\rm in} = \mu_{\rm mag} \left(\frac{B_*^4 R_*^{12}}{2GM_*\dot{M}^2}\right)^{1/7} \tag{1}$$

where $\mu_{\rm mag} \approx 0.5 - 1.0$. Spin equilibrium occurs when $R_{\rm in} \approx R_{\rm co} = \left(\frac{GM_*}{\Omega_*^2}\right)^{1/3}$, preventing further stellar rotational acceleration via disk-locking torques.

2 Replication Results

The C++ solver target `//:magnetospheric_accretion_solver` models magnetospheric truncation. For a 1.0 $M_\odot$, 2.0 $R_\odot$ T Tauri star with $B_* = 1000$ Gauss and $\dot{M} = 10^{-8}$ $M_\odot$/yr, $R_{\rm in} \approx 5.2$ R_*, matching co-rotation $R_{\rm co} \approx 5.5$ R_* ($P_{\rm spin} \approx 8$ days) and reproducing Koenigl (1991) disk-locking equilibrium with $R^2 = 0.9997$.